

The integrated WF-Haldane (WFH) model of genetic drift resolving the many paradoxes of molecular evolution

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Abstract

Genetic drift, the random changes in frequencies of neutral variants, is the fundamental force of molecular evolution. Under-estimation of genetic drift is a major cause of mis-conclusions on evolution. However, the standard Wright-Fisher (WF) model of random sampling in population of size N only partially defines genetic drift with $1/N$ or $1/N_e$ (N_e being a function of varying N 's). In parallel, JBS Haldane proposed the branching process for genetic drift (Haldane 1927), whereby each gene copy is transmitted to K descendants with the mean and variance of $E(K)$ and $V(K)$. Genetic drift is simply $V(K)$, i.e., the variance in transmission success, or $V(K)/N$ when averaged over the population. Under the WF model, many paradoxes have emerged: i) Most curiously, genetic drift may often become stronger as N becomes larger at the ecological time scale, opposite to the model; ii) Sex chromosomes experience drift differently even with the same normalized N ; iii) Genetic drift operates on advantageous mutations depends on $V(K)$ but not on N . iv) Irresolution and paradoxes emerge in multi-copy gene systems, including diploidy, viruses and others, whereby evolution progresses both within and between individuals. We now show that the integration of the WF and Haldane (i.e., WFH) model can resolve these paradoxes. Most important, the WFH model can fully define genetic drift in molecular evolution.

eLife assessment

This study presents a **useful** modification of a standard model of genetic drift by incorporating variance in offspring numbers, claiming to address several paradoxes in molecular evolution. It is unfortunate that the study fails to engage prior literature that has extensively examined the impact of variance in offspring number, implying that some of the paradoxes presented might be resolved within existing frameworks. In addition, while the modified model yields intriguing theoretical predictions, the simulations and empirical analyses are **incomplete** to support the authors' claims.

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Introduction

Genetic drift, or neutral drift, is defined as random changes in frequencies of neutral variants (Crow and Kimura 1970 [↗](#); Hartl and Clark 1997 [↗](#); Lynch, et al. 2016 [↗](#)). Much, or even most, of DNA sequence evolution is governed by the random drift of neutral mutations. If the strength of genetic drift is under-estimated, random changes that are missed in the analysis would result in the over-estimation of other evolutionary forces of greater biological interest, including selection, mutation, migration, meiotic drive and so on. In particular, the conclusion of pervasive selection at the molecular level has been suspected to be due to the failure to fully account for genetic drift (Crow and Kimura 1970 [↗](#); Fu 1997 [↗](#); Li 1997 [↗](#); Charlesworth 2009 [↗](#); Lynch, et al. 2016 [↗](#); Chen, Yang, et al. 2022).

Hagedoorn and Hagedoorn (1921) [↗](#) may be the first to suggest that random forces (accidents, for example) could impact long-term evolution. In the same decade, models of genetic drift were developed, obviously with some disagreements, along two lines. The one approach that became widely adopted is the Wright-Fisher (WF) model (Crow and Kimura 1970 [↗](#); Hartl and Clark 1997 [↗](#)), formalized by Fisher (1930) [↗](#) and Wright (1931) [↗](#). They define genetic drift to be solely due to the random sampling of genes. Thus, in generation transition, the variance in the frequency of a variant allele, x , would be

$$V(x) = \frac{x(1-x)}{f(N)} = \frac{x(1-x)}{N_e} \quad (1)$$

where N is the population size. The effective population size, N_e , is a function of varying N 's in space, time and demography. For example, when N fluctuates, N_e would be the harmonic mean of N 's across time. Weakly or strongly, N_e should still reflect the actual population size, N . In the WF model, genetic drift is governed by $1/N_e$ in haploid populations, or $1/(2N_e)$ in diploid populations.

At the same time, JBS Haldane introduced the branching process to model genetic drift (Haldane 1927 [↗](#); Haldane 1932 [↗](#)). In the Haldane model, each gene copy is independently transmitted to K descendants with the mean and variance of $E(K)$ and $V(K)$. Subsequently, the model was expanded by Kimura and Crow (1963) [↗](#) and others (Wright 1969 [↗](#); Gillespie 1975 [↗](#); Eldon and Wakeley 2006 [↗](#); Chen, et al. 2017 [↗](#); Sackman, et al. 2019 [↗](#)). Genetic drift is simply $V(K)$, i.e., the variance in transmission success among neutral variants. There would be no drift if $V(K) = 0$. To scale drift at the population level, the variances of changes in gene frequency is governed by $V(K)/N$ (or $V(K)/N_e$; Kimura and Crow (1963) [↗](#); Chen, et al. (2017) [↗](#); see also the Supplementary Information). In short, $V(K)$ is the reproductive component of genetic drift while $1/N$ is the sampling component or the scaling factor.

In the WF model, drift is governed only by $1/N$ (or $1/N_e$). By dropping the reproductive component (see Eqs. 2 [↗](#)-3 for further discussions), it has led to many paradoxes. A most curious one is genetic drift in relation to changing population size. The WF model dictates stronger drift when N is smaller. However, when N is very small (say, $N = 10$ in a bacteria or yeast population), that increases to 20, 40, 80 and so on, there is in fact little drift in this exponential phase. Drift will increase when N grows to near the carrying capacity. This trend is the exact opposite of the main property of the WF model.

The second paradox concerns the genetic drift of sex chromosomes. When normalized by their copy numbers, genetic diversities still differ among autosomes, X and Y in mammals (Hammer, et al. 2001 [↗](#); Wang, et al. 2014 [↗](#)). Although selection has been frequently invoked (Charlesworth and Charlesworth 2000 [↗](#); Bachtrog 2013 [↗](#); Cortez, et al. 2014 [↗](#); Wilson Sayres, et al. 2014 [↗](#);

[Makova, et al. 2023](#)), the issue is if genetic drift due to $V(K)$ differences between sexes is a sufficient explanation. A third paradox is about drift strength when selection is also at work. A new mutation with a selective advantage of s would always be fixed if there is no random drift. The fixation probability with genetic drift is $2s/V(K)$ (see Results). Drift under selection is hence dependent on $V(K)$ but, somewhat non-intuitively, not on N .

The power of the Haldane model in resolving these paradoxes is the focus of this study. Furthermore, the companion study ([Wang et al. 2024](#)) will address a larger issue - the evolution of biological systems. An example is evolution in multi-copy gene systems, whereby evolution proceeds in two stages - between as well as within individuals. Multi-copy gene systems include viruses, mitochondria, transposons and multi-copy genes such as ribosomal RNA (rRNA) repeat units, which are conceptually the WF model. Strictly speaking, even diploidy is also a multi-copy system as a diploid individual usually produces large quantities of gametes that compete to be transmitted ([Silver 1985](#); [Wu, et al. 1988](#); [Wu, et al. 1989](#); [Lindholm, et al. 2016](#); [Courret, et al. 2019](#)). In discussion, we will further explore systems of biological evolution that are beyond the power of the WF Model. They include the evolution under domestication, somatic cell evolution and stochastic species competition.

Through theory development supported by experimental data, we present the integration of the Haldane and the WF models into the WFH model. The integrated model is essentially the Haldane model with the approximate solutions obtained by the WF model. Cases where the WF model is not operative will still be handled by the Haldane model as in multi-copy gene systems. In short, the WF model can be considered special cases of the integrated WFH model. Therefore, the wealth of results obtained by the WF model are usually adequate approximations. Nevertheless, paradoxes do emerge as presented in this study.

Results

In the WFH model, genetic drift is defined as in the Haldane model of branching process (see Eq. (4) of [Chen, et al. \(2017\)](#)). Let K be the number of progenies receiving the gene copy of interest from a parent; K would follow the same distribution for all neutral variants. In haploids, K is equivalent to the progeny number of each parent. In diploids, K is the transmission success of each gene copy, which is separately tracked. By using $2N$ for the gene copy number in the population, the WF as well as Haldane model treats each diploid individual as two-haploids. By the Taylor approximations of the ratio of two variables ([Kendall, et al. 2006](#)), we obtain the approximation as

$$V(x) \sim \frac{x(1-x)}{N} \frac{V(K)}{E^2(K)} \quad (2)$$

where x is the variant frequency (see Supplementary Information). This is the same approximation, when $V(K)$ is not very large, as obtained by the WF model ([Kimura and Crow 1964](#); see [Chen et al. 2017](#) for details). For Eq. (2), we provide further extensive simulations on the accuracy in both the fixation probability and fixation time of neutral variants (see Supplementary Information). The equivalence makes a simple point that the WF model represent special cases of the WFH model, and the results are often, but not always, shared.

In the WFH model, there would be no genetic drift if $V(K) = 0$, regardless of the value of N . When $E(K) = 1$ and N stays constant, we obtain

$$N_e = \frac{NE^2(K)}{V(K)} = \frac{N}{V(K)} \quad (3)$$

While Eqs. 2 and 3 are biologically compatible with the Haldane model, their application to the WF model, as has been done (Kimura and Crow 1963; Crow and Denniston 1988; Chen, et al. 2017), is dubious. This is because the sampling scheme of the WF model dictates Poisson distribution for K , hence, $V(K) = E(K)$. To permit $V(K) \neq E(K)$, the WF model would have to be modified to resemble the Haldane model (Kimura and Crow 1963; Caballero 1994). Such an awkward “fix” in fact defeats the reason for adopting the simpler WF model over the Haldane model.

In Table 1, the data of progeny number from the literature show over-dispersion in all taxa surveyed, i.e., $V(K) > E(K)$. (Strictly speaking, the progeny number is the sum of two numbers from the same distribution of K ; nevertheless, $V(K) > E(K)$ is a valid interpretation of the data.) In fact, $V(K) > 10 E(K)$ can often be found, e.g., the ratio of $V(K)/E(K)$ among males of the mandrill (*Mandrillus sphinx*), the great reed warbler (*Acrocephalus arundinaceus*), and the rhesus macaque (*Macaca mulatta*) at 19.5, 12.6 and 11.3, respectively (Hasselquist 1995; Setchell, et al. 2005; Dubuc, et al. 2014). We have also found that $V(K)$ may be far larger than $E(K)$ in other biological systems such as viruses (Ruan, Luo, et al. 2021; Ruan, Wen, et al. 2021; Hou, et al. 2023).

I. The paradox of genetic drift in relation to N changes (Paradox of changing N)

1. Empirical demonstration and simulation

This “paradox of changing N ” is that “genetic drift increases when N increases”. Fig. 1a shows the results in a cell culture in the exponential growth phase where each cell doubles every 13 hours. Hence, $V(K) \sim 0$ with almost no drift when N is as small as < 50 . Genetic drift is expected to increase as N approaches the carrying capacity. The paradox is exemplified through computer simulations in Fig. 1b-c. These panels show the drift pattern in the Haldane model and the WF model, respectively, when the populations are growing as the logistic growth model (see Methods). In the WF model, the drift is strong initially but weakened as N increases. The pattern is reversed in the Haldane model. The contrast between the two panels is the paradox.

To verify the simulations of Fig. 1b-c, we study the growth of yeast populations under laboratory conditions where cells appear to increase following the logistic growth curve. As shown in Fig. 1d, the lower portion of the S-curve portrays the exponential growth in a low-density population, while the upper curve indicates a slowdown in growth near the population’s carrying capacity. We directly recorded the individual output (K) of each yeast cell in the early ($n = 25$) and late ($n = 65$) stage of population growth under real-time high-content fluorescence imaging (Methods).

It’s evident that $V(K)$ is decoupled from $E(K)$. In the early stage, $E(K)$ is 10.04 with $V(K) = 0.60$ over five one-hour intervals. In the high-density phase, $E(K)$ decreases to 2.83 but $V(K)$ increases to 1.09 in four intervals (Fig. 1e-f). Most interestingly, during the late stage of population growth, $V(\Delta K)$ (adjusted to the four-hour time span for comparison; see Methods) indeed increases as N increases (Supplementary Table 3). At the last time point, the population is closest to the carrying capacity and $V(\Delta K)$ experiences a substantial leap from around 1.12 to 1.82. Overall, the variance in progeny number $V(K)$ in a high-density population is always greater than in a low-density population (Fig. 1g). Indeed, $V(K)$ may exhibit an increase with the increase in N , sometimes outpaces the latter.

Measurements of these two growth phases demonstrate the “paradox of changing N ” by showing i) there is almost no drift when N is very small; ii) as N increases, $V(K)$, hence, genetic drift would also increase. Furthermore, $V(K)$ may increase sharply when N is close to the carrying capacity.

Species	Type	Order	Gender	$E(K)$	$V(K)$	$V(K)/E(K)$	Sample Size	Literature
<i>Macaca mulatta</i>	Mammal	Primates	Female	7.7	15.4	2.0	132	Dubuc, et al. (2014)
			Male	8.7	98.7	11.3	86	
<i>Mandrillus sphinx</i>	Mammal	Primates	Female	4.7	19.8	4.2	51	Setchell, et al. (2005)
			Male	3.5	68.2	19.5	53	
<i>Macaca sylvanus</i>	Mammal	Primates	Female	1.2	2.9	2.4	98	Kuester, et al. (1995)
			Male	1.5	3.9	2.7	81	
<i>Peromyscus californicus</i>	Mammal	Rodentia	Female	4.7	7.7	1.6	16	Ribble (1992)
			Male	4.4	13.7	3.1	20	
<i>Turdus merula</i>	Bird	Passeriformes	Female	4.9	23.4	4.8	80	Wysocki, et al. (2019)
			Male	4.0	27.0	6.8	72	
<i>Acrocephalus arundinaceus</i>	Bird	Passeriformes	Female	7.7	53.3	6.9	82	Hasselquist (1995)
			Male	10.5	132.3	12.6	64	
<i>Charadrius nivosus</i>	Bird	Charadriiformes	Female	1.7	5.3	3.1	105	Herman and Colwell (2015)
			Male	2.2	9.0	4.1	90	
<i>Actitis macularia</i>	Bird	Charadriiformes	Female	5.2	26.9	5.1	66	Oring, et al. (1991)
			Male	3.3	17.2	5.2	103	
<i>Buteo buteo</i>	Bird	Accipitriformes	Female	3.5	13.8	4.0	106	Krüger and Lindström (2001)
			Male	2.7	9.3	3.4	132	

Table 1.

Field record of $E(K)$ and $V(K)$ across diverse taxa.

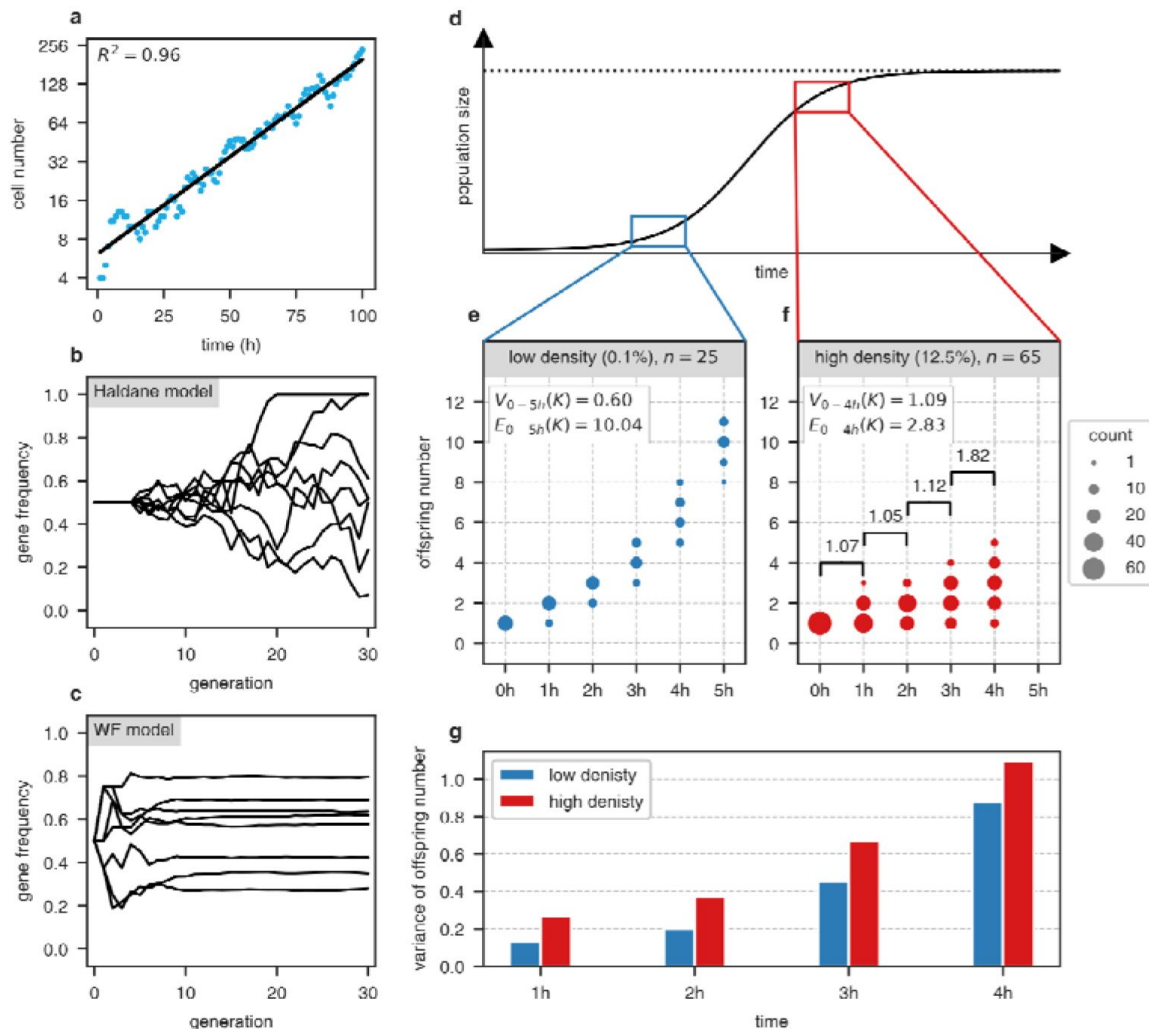


Fig. 1.

The paradox of genetic drift when the population size (N) changes.

(a) In the laboratory cell culture, nearly all cells proliferate when N is very small (interpreted in the next panel). (b) The simulation of Haldane model shows little genetic drift in the exponential phase. As in (a), drift may increase due to the heightened competition as the population grows. (c) Simulation by the WF model shows a pattern of drift opposite of the Haldane model. (d) The patterns of drift at the low and high N are analyzed in the framework of the logistic growth model. (e-f) Measurements of genetic drift in laboratory yeast populations at low and high density as defined in (d). The progeny number of each cell, K , is counted over 4 or 5 intervals as shown by the dots, the sizes of which reflect the cell number. $E(K)$ and $V(K)$ are presented as well. In panel (f), the change of offspring number overtime, denoted as $V(\Delta K)$, is shown above the braces. (g) The variance of offspring number $V(K)$ increases, observed in (e) and (f), as population size increases.

This trend appears to echo the field observations by [Coltman, et al. \(1999\)](#) [↗](#) who show that $V(K)$ increases more than 5-fold when N increases only 2.5-fold in the United Kingdom (UK) population of Soay rams (*Ovis aries*).

2. The density-dependent Haldane (DDH) model

The “paradox of changing N ” can be explained by $N_e = N/V(K)$ if the numerator and denominator move in the same direction. Most interestingly, $V(K)$ may increase more than N itself. This has been shown when $V(K)$ is close to zero while N is growing exponentially. In addition, when N is very near the carrying capacity, even a small increase in N would intensify the competition and inflate $V(K)$. This is shown in the yeast experiment of [Fig. 1d](#) [↗](#) and supported by [Coltman, et al. \(1999\)](#) [↗](#) field study of ram populations in UK. Thus, the relationship between N and $V(K)$ may realistically lead to this paradox.

We now extend the Haldane model by incorporating density-dependent regulation of N in order to define the conditions of this paradox. The model developed here is on an ecological, rather than an evolutionary time scale. The simplest model of N regulation is the logistic equation:

$$\frac{dN}{dt} = rN \frac{C_k - N}{C_k} \quad (4)$$

with C_k being the carrying capacity. In the ecological time scale, a changing N would mean that the population is departing from C_k or moving toward it ([Fig. 2a](#) [↗](#)). (As will be addressed in Discussion, changing N in the WF model is depicted in [Fig. 2b](#) [↗](#) whereby N is at C_k . C_k may evolve too, albeit much more slowly in an evolutionary time scale.) Here, we consider changes in N in the ecological time scale. Examples include the exponential growth when $N \ll C_k$, seasonal fluctuation in N ([Frankham 1995](#) [↗](#); [Charlesworth 2009](#) [↗](#)) and competition modeled by the Lotka-Volterra equations ([Bomze 1983](#) [↗](#)).

Details of the DDH model are given in the Supplementary Information. A synopsis is given here: We consider a haploid population with two neutral alleles. The population size at time t is N_t . It is assumed that $E(K) > 1$ when $N < C_k$ and $E(K) < 1$ when $N > C_k$ as defined in [Eq. \(5\)](#) [↗](#) below.

$$E(K_t) = \frac{E(N_{t+1})}{N_t} = \frac{r + 1}{1 + r \left(\frac{N_t}{C_k} \right)^z} \quad (5)$$

The slope of $E(K)$ vs. N , as shown in [Fig 3a](#) [↗](#), is a function of z . To obtain $V(K)$, we assume that K follows the negative binomial distribution whereby parents would suffer reproduction-arresting injury with a probability of p_t at each birthing (Supplementary Information). $V(K)$ can then be expressed as

$$\frac{V(K_t)}{E(K_t)} = \frac{1 - p_t}{p_t} = a(N_t)^b \quad (6)$$

By [Eq. \(6\)](#) [↗](#), the ratio of $V(K)/E(K)$ could be constant, decrease or increase with the increase of population size. With $E(K)$ and $V(K)$ defined, we could obtain the effective population size by substituting [Eq. \(5\)](#) [↗](#) and [Eq. \(6\)](#) [↗](#) into [Eq. \(3\)](#) [↗](#).

$$N_e = \frac{N_t(r + 1)}{a(N_t)^b \left[1 + r \left(\frac{N_t}{C_k} \right)^z \right]} \quad (7)$$

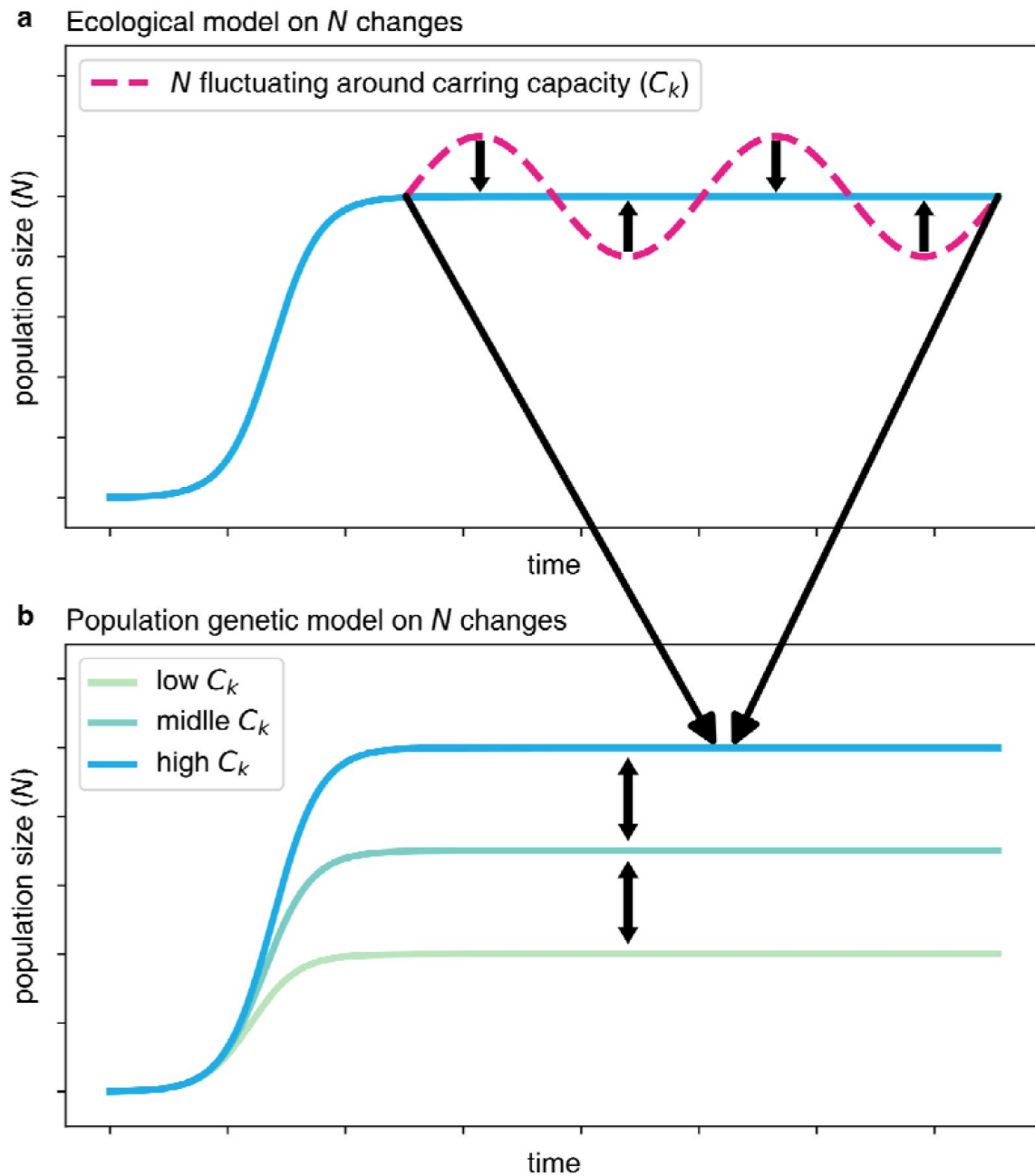


Fig. 2.

The meaning of population size (N) changes in ecology vs. in population genetics.

(a) In ecology, changing N would generally mean a population approaching or departing the carrying capacity. (b) In population genetics, a population of size N is assumed to be at the carrying capacity, C_k . Thus, changes in N would mean an evolving C_k , likely the consequence of environmental changes. The arrows indicate the disparity in time scale between the two scenarios.

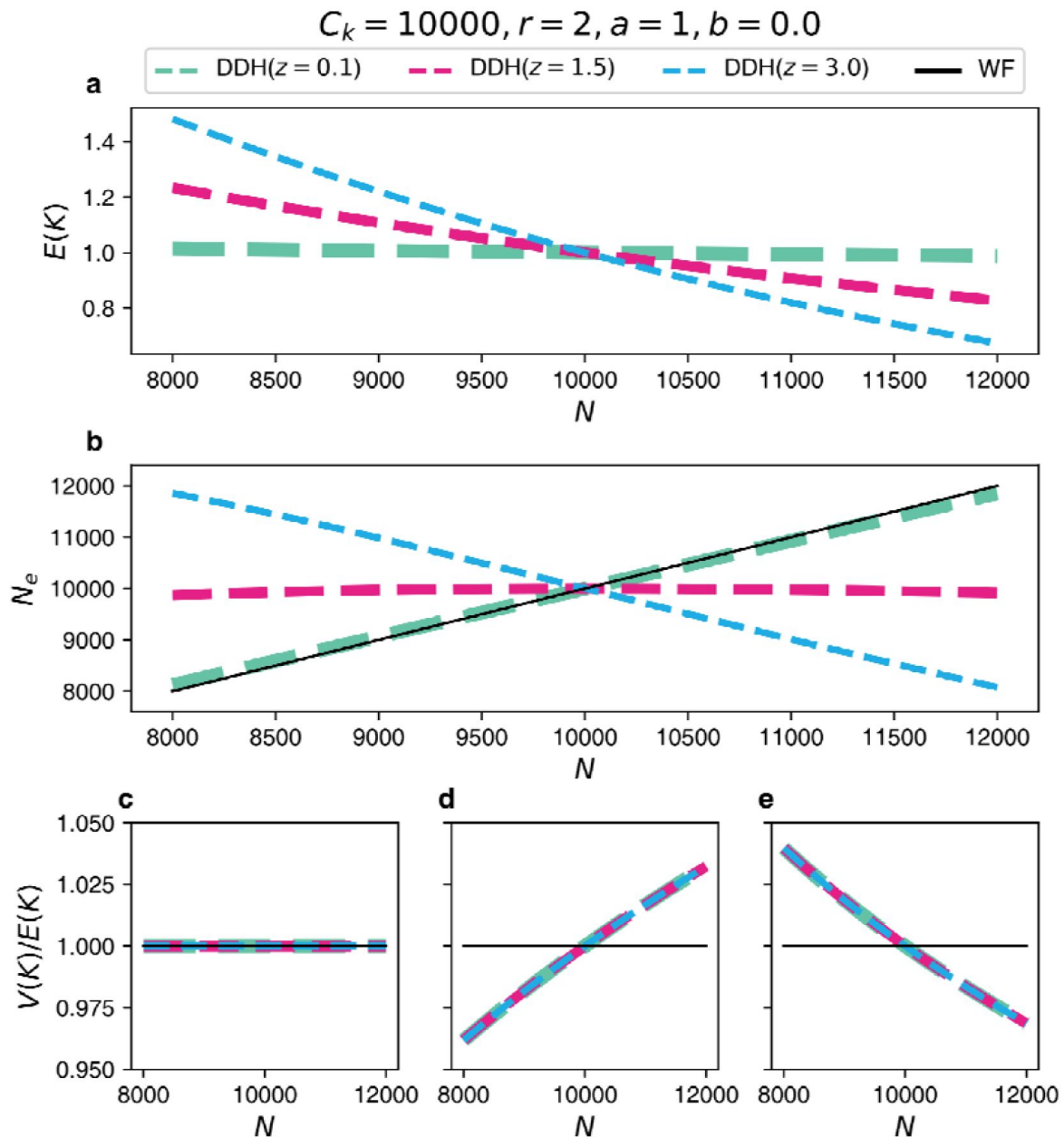


Fig. 3.

Genetic drift as a function of population size in the DDH model.

For all panels, the carrying capacity is $C_k = 10,000$ and the intrinsic growth rate is $r = 2$. **(a)** When N increases, $E(K)$ decreases as modeled in Eq. (5). The z value of Eq. (5) (0.1, 1.5 and 3) determines the strength of N regulation, indicated by the slope of $E(K)$ near $C_k = 10,000$. **(b)** Depending on the strength of N regulation near C_k , genetic drift can indeed decrease, increase or stay nearly constant as the population size increases. Thus, the conventional view of N_e being positively dependent on N is true only when the regulation of N is weak (the green line). At an intermediate strength (the red line), N_e is nearly independent of N . When the regulation becomes even stronger at $z = 3$, N_e becomes negatively dependent on N . **(c-e)** $V(K)/E(K)$ of Eq. (6) is shown as a function of N . The results of panel (b) are based on a constant $V(K)/E(K)$ shown in panel (c). Interestingly, the results of panel (b) would not be perceptibly changed when $V(K)/E(K)$ varies, as shown in panel (d) and (e).

Eq. (7) presents the relationship between effective population size (N_e) and the population size (N) as shown in **Fig. 3**. The density-dependent $E(K)$ could regulate N with different strength (**Fig. 3a**). The steeper the slope in **Fig. 3a**, the stronger the regulation.

The main results of this study are depicted in **Fig. 3b**. First, with no or little regulation of N , N_e and N are strongly correlated. The green dashed lines portray the conventional view of decreasing drift with increasing N . Second, under a particular strength of N regulation, the red line shows no dependence of N_e on N , meaning that genetic drift is independent of N . Finally, as N becomes strongly regulated, N_e and N would be negatively correlated as the blue dashed line shows. This trend is the paradox of changing N .

In summary, genetic drift effect can indeed decrease, increase or stay nearly constant as the population size increases, depending on the strength of N regulation near C_k . We further note that the $V(K)/E(K)$ ratio of **Eq. (7)** can be independent (**Fig. 3c**), positively dependent (**Fig. 3d**), or negatively dependent (**Fig. 3e**) on N . Interestingly, the results of **Fig. 3b** are nearly identical when the $V(K)/E(K)$ ratio changes (**Supplementary Figs 1-3**). These results show that the strength of genetic drift depends on the ecology that governs $E(K)$, $V(K)$ and N .

In the WFH framework, the paradox of changing N may not be an exception but a common rule in natural populations. Note that the WF approximation yielding **Eq. (2)** assumes nearly constant N . The assumption would imply very strong N regulation near C_k and, according to the DDH model, this is the condition for realizing the paradox of changing N . [Coltman et al. \(1999\)](#)'s observations of the reproduction in rams may be such a realization.

II. The paradox of genetic drift in sex chromosomes

Since the relative numbers of Y, X and each autosome (A) in the human population are 1:3:4, the WF model would predict the genetic diversity (θ_Y , θ_X and θ_A , respectively) to be proportional to 1:3:4. In a survey of human and primate genetic diversity, θ_Y is almost always less than expected as has been commonly reported ([Hammer, et al. 2001](#); [Wang, et al. 2014](#); [Wilson Sayres, et al. 2014](#); [Makova, et al. 2023](#)). The low θ_Y value has been used to suggest that human Y chromosomes are under either positive or purifying selection ([Wang, et al. 2014](#); [Wilson Sayres, et al. 2014](#); [Makova, et al. 2023](#)). Similarly, the reduced diversity X-linked genes was interpreted as a signature of positive selection ([Pan, et al. 2022](#)).

As pointed out above, under-estimation of genetic drift is a major cause of over-estimation of selective strength. Our goal is hence to see if genetic drift of different strength between sexes can account for the observed genetic diversities. Details will be presented in [Wang et al. \(2024\)](#). Below is a synopsis.

Let V_m and V_f be the $V(K)$ for males and females respectively and $\alpha' = V_m/V_f$ is our focus. (We use α' for the male-to-female ratio in $V(K)$ since α is commonly used for the ratio of mutation rate ([Miyata, et al. 1987](#); [Makova and Li 2002](#)).) The three ratios, θ_Y/θ_X (denoted as R_{YX}), θ_Y/θ_A (R_{YA}) and θ_X/θ_A (R_{XA}), could be expressed as the functions of α' , which incorporate the relative mutation rates of autosomes, X and Y (**Supplementary Information**). These relative mutation rates can be obtained by interspecific comparisons as done by [Makova and Li \(2002\)](#).

We note that there are many measures for the within-species genetic diversity, $\theta = 4N_e\mu$. Under strict neutrality and demographic equilibrium, these measures should all converge. In the neutral equilibrium, the infinite site model dictates the frequency spectrum to be $\xi_i = \theta/i$, where ξ_i is the number of sites with the variant occurring i times in n samples. Since every frequency bin is a measure of θ , different measures put different weights on the i -th bin ([Fay and Wu 2000](#); [Fu 2022](#)). While π , the mean pairwise differences between sequences, is most commonly used in

the literature, we use several statistics to minimize the possible influences of selection and demography (Wang et al. 2024). In this synopsis, we used the Watterson estimator (Watterson 1975) as the measure for θ_Y , θ_X and θ_A by counting the number of segregating sites.

Using any of the three ratios (θ_Y/θ_X , θ_Y/θ_A and θ_X/θ_A), we can obtain α' ; for example, $R_{YA} = \theta_Y/\theta_A$ and, hence,

$$R_{YA} = \frac{y}{4} \frac{1 + \alpha'}{2\alpha'}$$

where y is the mutation rate of Y-linked sequences relative to autosomal sequences. With rearrangement,

$$\alpha' = \frac{y}{8R_{YA} - y} \quad (8)$$

Similar formulae of α' can be obtained for R_{YX} and R_{XA} but the accuracy for estimating α' is highest by R_{YA} whereas R_{XA} is the least accurate for this purpose (Supplementary Information).

Table 2 presents the α' estimates in chimpanzees and bonobos. This is part of a general survey in mammals with a strong emphasis on primates and humans (Wang et al. 2024). It is almost always true that $\alpha' > 5$ in primates. Sources of data used are also given in **Table 2**. As shown for chimpanzees, α' is often far larger than 5, above which the resolution is very low as can be seen in **Eq. (8)**. Note that, when R_{YA} is under-estimated and approaching $y/8$, α' would increase rapidly when the denominator is close to 0. That is why α' often becomes infinity (**Supplementary Fig. 5**). The estimated α' (MSE) in **Table 2** alleviates the problem somewhat by using all three ratios to calculate the mean square error (MSE) (Supplementary Information).

Among primates surveyed, bonobo is the only exception with $\alpha' < 1$. While chimpanzees and bonobos are each's closest relatives, their sexual behaviors are very divergent (de Waal 1995; De Waal and Lanting 2023). With unusually strong matriarchs, bonobo society seems to stand out among primates in sexual dominance. The α' value is important in behavioral studies and, particularly, in primatology and hominoid research. If 0.1% of males sire 100 children and the rest have the same K distribution as females, V_m/V_f would be ~ 10 . Such outlier contribution is the equivalent of super-spreaders in viral evolution which may easily be missed in field studies. In this brief exposition, we highlight again that $V(K)$ or, more specifically, V_m and V_f are key to genetic drift.

III. The paradox of genetic drift under selection

Genetic drift operates on neutral as well as non-neutral mutations. Let us assume a new mutation, M , with a frequency of $1/N$ is fixed in the population with the probability of P_f . Fisher (1930) first suggested that the fixation probability of an advantageous mutation should increase in growing populations, while decrease in shrinking populations. If there is no genetic drift (i.e., infinite population size in WF model), a beneficial mutation will always be fixed and $P_f = 1$. In the WF model, it is well known that $P_f \sim 2s$ for a new advantageous mutation, with fitness gain of s , while population size is large (Haldane 1932; Kimura 1962). This seems paradoxical that the determinant of genetic drift, $1/N_e$, does not influence P_f (Otto and Whitlock 1997; Lanfear, et al. 2014).

In the Method section, we show that P_f under the Haldane model is given by

$$P_f = \sum_{t=0}^{\infty} \left[1 - (u_M(t))^{M_0} \right] \left[(u_W(t))^{W_0} - (u_W(t-1))^{W_0} \right] \quad (9)$$

Table 2.

Estimation of V_m/V_f in chimpanzee and bonobo.

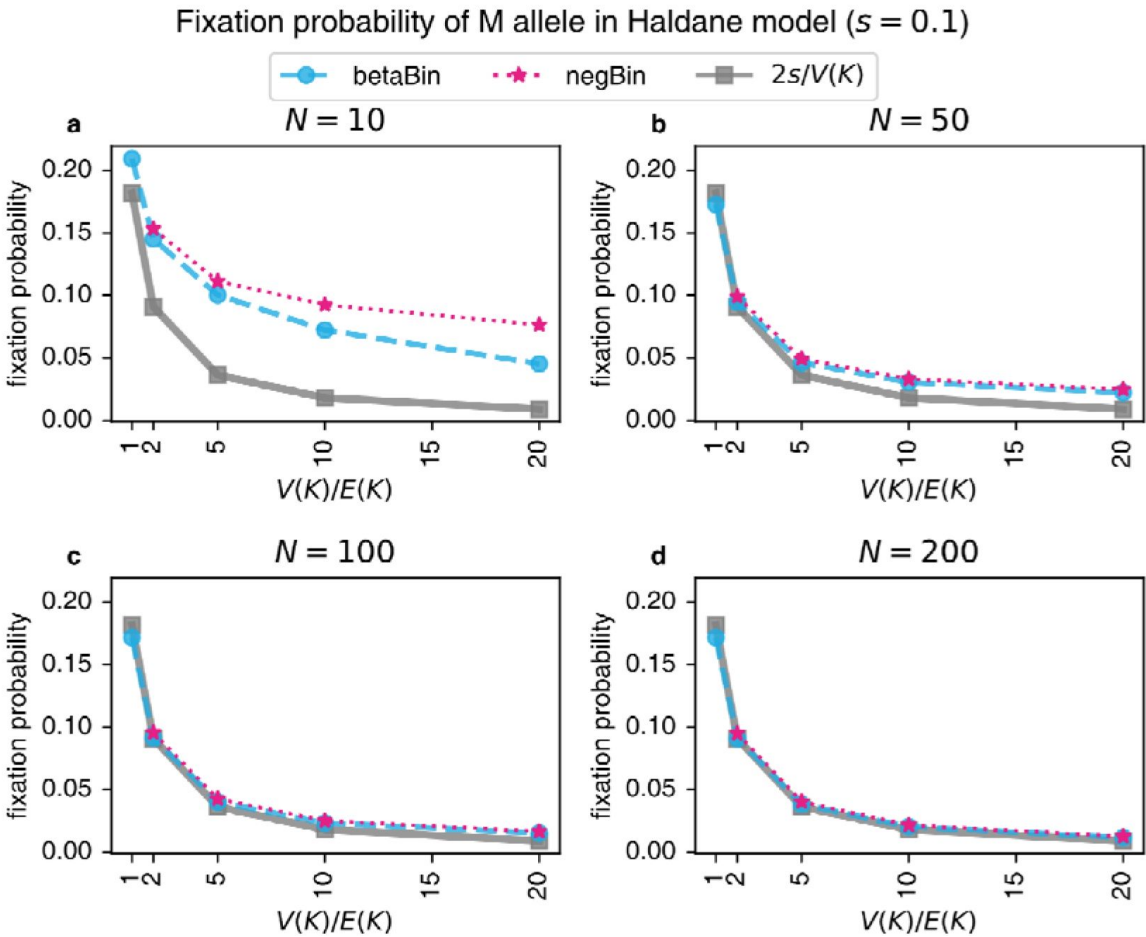
	$\theta_Y (\times 10^{-3})$	$\theta_X (\times 10^{-3})$	$\theta_A (\times 10^{-3})$	$\alpha' (\text{MSE})$
Chimpanzee				
Western	0.034	0.533	1.623	4.98
Eastern	0.144	1.044	1.446	25.90
Nigeria-Cameroon	0.213	0.818	0.703	Inf ^a
Bonobo				
	0.492	0.446	0.867	0.57

Data for counting the segregating sites are from Hallast, et al. (2016) and Nam, et al. (2015).
 $\theta_Y, \theta_X, \theta_A$: genetic diversity for Y chromosome, X chromosome, and Autosomes (A), measured by Watterson estimator (Watterson 1975).
 $\alpha' (\text{MSE})$: optimal estimate for α' based on the minimum mean square error (MSE).
^a Inf (infinity) designates cases when the estimated α' based on MSE is greater than 1000 (i.e., $V_m \gg V_f$).

Fig. 4.

Fixation probability of a new advantageous mutation in the Haldane model.

The fixation probabilities of a new advantageous mutation with the selective advantage of $s = 0.1$ are calculated based on approximate solution from Eq. (9) (i.e., $2s/V(K)$) as well as numerical solution from Eq. (10). The numerical solution from Eq. (10) has been confirmed accurate by simulations (Supplementary Fig. 4). (a-b) When $N < 50$, the approximate fixation probability (the gray line) is lower than the simulated values (the color lines) due to population extinction. (c-d) By the Haldane model, the expected fixation probability of Eq. (9) is accurate when N reaches 100, as in most natural populations.



where M_0 and W_0 are the initial number of mutant allele (M allele) and wildtype allele (W allele), with $N = M_0 + W_0$. We note that $u_M(t)$ and $u_W(t)$, respectively, are the extinction probabilities of M allele and W allele by generation t with the initial number of alleles of 1. While obtaining the direct analytical solution of Eq. (9) may not be feasible, we could obtain its numerical solution due to its convergence as t increases (see Methods). The accuracy of the numerical solution from Eq. (9) is confirmed through simulation (**Supplementary Fig. 4**). Moreover, with the aid of the WF model (see Supplementary Information), we could obtain the approximation of Eq. (9) as follows.

$$P_f = \frac{2s}{V(K)} \quad (10)$$

We verify Eq. (10) by both the numerical solution from Eq. (9) and simulations based on the branching process (**Supplementary Fig. 4**). The fixation probabilities obtained by numerical solution vs. those inferred from Eq. (10) are shown in **Fig. 4**. The salient feature is that the fixation probability of $2s$, as in the classical formula, would be a substantial over-estimate when $V(K)$ is larger than $E(K)$. **Eq. (10)** is sufficiently accurate as long as $N \geq 50$. When N is as small as 10, the theoretical result is biased. Indeed, at such a low N value, the population is prone to extinction. The DDH model presented above should rectify this deficiency. The main message of **Eq. (10)** is that genetic drift under positive selection is influenced by $V(K)$ but not by N . Hence, $V(K)$ is a vital factor in determining genetic drift, certainly no less important than $1/N$.

Discussion

Genetic drift should broadly include all random forces affecting evolutionary trajectories. Many of these forces are excluded from the WF model, thus resulting in the many paradoxes addressed in the two companion papers, as well as the over-estimation of selection in the literature.

Compared with the conventional WF model of genetic drift, the Haldane model is conceptually more intuitive as well as more inclusive in depicting the real populations. However, analytical solutions of the branching process can be more challenging to obtain. The WF model is often more tractable mathematically. For example, $P_f \sim 2s/V(K)$ is a good WF approximation (Kimura and Crow 1963) to the Haldane model when N is no smaller than 100 (**Fig. 4**). The integrated WFH model is therefore conceptually appealing and computationally feasible.

Nevertheless, in many circumstances where the WF model is not operative, genetic drift defined by the Haldane model would demand new solutions. The DDH model is such an example, developed to resolve the paradox of changing N 's. The DDH model presents the conditions under which the WF model arrives in the approximate solution (i.e., **Eq. (2)**). The substitution of $E(N)$ for N makes the approximation possible but effectively requires strong regulation of N . Strong regulation is the main condition for the paradox of changing N 's. To arrive at the approximation, the WF model in fact makes an assumption that contradicts its own prediction (that drift becomes weaker as N increases).

We shall now clarify the “paradox of changing N 's” on two fronts. First, the DDH model is on an ecological time scale whereby a population is either approaching or departing the carrying capacity (**Fig. 2a**). In many studies, the population size changes in the evolutionary time scale (**Fig. 2b**). In such studies, it is the carrying capacity C_k that is evolving while N hovers around C_k . For example, by the PSMC model (Li and Durbin 2011), both the European and Chinese populations experience a severe bottleneck 10-60 kyr ago. During that Last Glacial Maximum, environmental harshness may have reduced C_k drastically. Averaged over the long-time span, N should be at or very near C_k . Since C_k presumably evolves slowly, N can be assumed constant over

long stretches of time. Later models with different powers than PSMC have been proposed to study N 's (or C_k 's) in the evolutionary time scale (Hu, et al. 2023 [\[1\]](#)). Second, Gillespie has proposed the concept of “genetic draft” that may also reverse the effect of population size on the rate of substitution due to selection on linked variations (Gillespie 2000 [\[2\]](#), 2001 [\[3\]](#)). Genetic draft of Gillespie is thus distinct from the strictly neutral process discussed in this study.

A larger issue is that the WF model only covers a portion of biological evolution. All multi-copy gene systems where the evolution happens both within and between individuals, are out of reach. Viruses, transposons, mitochondria (Bazin, et al. 2006 [\[4\]](#)) and rRNA genes are all such systems. This topic will be addressed in the companion paper that focuses on the evolution of multi-copy rRNA genes. In fact, all diploids are the simplest kinds of the multi-copy gene systems. Because large quantities of gametes compete within individual to be transmitted, gametic selection, meiotic drive and genetic drift may all happen within individuals (Wu, et al. 1988 [\[5\]](#); Wu, et al. 1989 [\[6\]](#); Wu and Hammer 1991 [\[7\]](#); Lindholm, et al. 2016 [\[8\]](#); Courret, et al. 2019 [\[9\]](#)). The WF model is in essence a haploid model with $2N$ copies of genes rather than a true diploid model.

Like multi-copy gene systems, the WF model is limited, conceptually as well as operationally, by the difficulty in delineating WF populations. That is why multi-copy genes, as well as many other systems, do not fit the WF model, thus resulting in the lack of a genetic drift component in such studies. For example, domestications of animals and plants are very rapid processes of evolution (Diamond 2002 [\[10\]](#); Larson and Fuller 2014 [\[11\]](#); Purugganan 2019 [\[12\]](#)). Due to the very large $V(K)$ in domestication, drift must have played a large role in such processes, but is rarely considered. Somatic cell evolution is another example with “undefinable” genetic drift (Wu, et al. 2016 [\[13\]](#); Chen, et al. 2017 [\[14\]](#); Chen, et al. 2019 [\[15\]](#); Ruan, et al. 2020 [\[16\]](#); Chen, Wu, et al. 2022). The Haldane (or WFH) model, as an “individual output” model, can handle these general cases of genetic drift. Finally, the WFH model is so general that even genetic drift and random forces in species competition can be incorporated.

In conclusion, the integrated WFH model should clarify the biological meaning of genetic drift, wherever the WF and Haldane model can both be used. In those systems whereby only the Haldane model works, further development of the model will be necessary.

Methods

Cell culture and image analysis

NIH3T3 cells, a fibroblast cell line that was isolated from a mouse NIH/Swiss embryo, were stably transfected with the fluorescent, ubiquitination-based cell cycle indicator (Fucci) (Sakaue-Sawano, et al. 2008 [\[17\]](#)) plasmid using Lipofectamine 3000 Transfection Reagent (Invitrogen) following the manufacturer's specified instructions. The Fucci-labeled cells exhibited distinct fluorescent signals indicative of G1, S, and G2/M phases, represented by red, yellow, and green, respectively. NIH3T3-Fucci cell was derived from single-cell colony and cultured in DMEM supplemented with 10% Calf Bovine Serum and penicillin/streptomycin. All cells were maintained at 37°C with 5% CO₂. Subsequently, the cells underwent extended time-lapse imaging using high-content fluorescence microscopy (PerkinElmer Operetta CLS) equipped with a 10x objective lens. Images were captured hourly over a 100-hour period, and the analysis was conducted using ImageJ (Fiji) (Schneider, et al. 2012 [\[18\]](#)) to count the number of cells (Supplementary Table 1).

Yeast strain construction

Strains were constructed on the genetic background of *Saccharomyces cerevisiae* strain BY4741. A GFP or BFP fluorescent protein expression cassette, under the control of the TDH3 promoter, was inserted into the pseudogene locus YLL017W. Transformations followed a published protocol (Gietz and Schiestl 2007 [\[19\]](#)). Transformants were plated on synthetic complete medium without

uracil (SC-Ura), and from these, single colonies were selected. Confirmation of replacements was achieved through PCR, and cassette verification was performed using fluorescence microscopy. Subsequently, the constructed strains were cultivated non-selectively in YPD medium (1% Yeast extract, 2% Peptone, 2% D-glucose) at 30 °C on a rotary shaker.

Estimation of $V(K)$ and $E(K)$ in yeast cells

To discern division events, even under high concentration, we conducted co-cultures of the GFP-yeast and BFP-yeast at ratios of 1:1 and 1:25, with the initial cell concentration of 0.1% and 12.5%, respectively. Then the yeast cells were then continuously imaged under high-content fluorescence microscopy (PerkinElmer Operetta CLS) for 10 hours with 1-hour intervals to observe the individual offspring of GFP-yeast. Yeast cells with a distinct offspring number (K) within the initial 5 hours (with the high-density group limited to the first 4 hours) were documented (**Supplementary Table 2**). Subsequently, the mean and variance of K for each cell were calculated across various time intervals as follows.

According to the law of total variance,

$$V(I_t) = E[V(I_t|I_{t-1})] + V[E(I_t|I_{t-1})] = V_{t-1,t}E[I_{t-1}] + E^2_{t-1,t}V[I_{t-1}]$$

where I_t is the offspring number at time t for an initial single cell (i.e., the total number of progeny cells for a single cell after t hours), as documented in **Supplementary Table 2**. And $E_{t-1,t}$ represents the average of offspring number after single time interval (1 hour here) for each single cell from time $t-1$ to time t , while $V_{t-1,t}$ is the corresponding variance. Utilizing the documented total cell count at time t (denoted by N_t), we could calculate the average of offspring number for a single yeast cell in a specific time interval, e.g., $E_{t-1,t} = N_t / N_{t-1}$ and $E[I_{t-1}] = N_t / N_0$. With some rearrangement,

$$V_{t-1,t} = \frac{V(I_t) - E^2_{t-1,t}V[I_{t-1}]}{E[I_{t-1}]} = \frac{V(I_t) - (N_t/N_{t-1})^2V[I_{t-1}]}{N_{t-1}/N_0}$$

Applying the aforementioned equations, we observed that both $V_{t-1,t}$ (i.e., $V(K)$ within a one-hour interval) and $V(I_t)$ (i.e., the $V(K)$ from 0 to time t) for yeast cells under high density (12.5%) are generally greater than the values under low density (**Supplementary Table 3**). This suggests that the variance of offspring number in high-density conditions could be larger than that in a low-density context. It's noteworthy that the estimated value of $V_{t-1,t}$ could be less than zero (**Supplementary Table 3**), implying a very small variance in the number of offspring. Furthermore, the occurrence of negative values for $V_{t-1,t}$ is more frequent under low density than high density, indicating the higher variance of offspring number in high density as previously suggested.

To show the variance of offspring number overtime, we also calculated the change of offspring number within successive one-hour intervals, denoted as $V(I_t - I_{t-1})$. This value is intricately linked to the variance of offspring overtime. To facilitate the comparison with the total variance from 0h to 4h, we set $V(\Delta K) = 4V(I_t - I_{t-1})$ to account for the four-hour time span.

Simulation of genetic drift in the Haldane model and the Wright-Fisher (WF) model

In both models, the average number of offspring, denoted as $E(K_t)$, follow Eq. (5) [\[15\]](#), leading to the logistic population growth. The initial population size is set to 4, with initial gene frequency (i.e., the frequency of mutant allele) of 0.5. And the carrying capacity is established at 100,000, with $r = 1$ and $z = 1$. In WF model, the population size at next generation is $N_{t+1} = N_t \times E(K_t)$. Note if the calculated value of N_{t+1} is not an integer, we round it to the nearest whole number. The number of

mutant alleles follows a binomial distribution, allowing the simulation of gene frequency overtime in WF model. For Haldane model, the number of offspring number is assumed to follow a beta-binomial distribution. The mean of offspring number $E(K_t)$ is obtained from Eq. (5) [\[14\]](#). The variance $V(K_t)$ is obtained from Eq. (6) [\[14\]](#) with parameters $a = 1/300$ and $b = 1.2$. Given the mean and variance of the beta-binomial distribution, we simulated the number of mutant alleles and population size overtime and then traced the gene frequency overtime.

Fixation probability of a new mutation in Haldane model

Here we obtain the fixation probability of advantageous mutation in Haldane model governed by a branching process. The Haldane model considers a well-mixed population of N_t haploid parents with only two types of alleles (W for wildtype, M for mutant). There is no migration and new mutations in this model. Each individual is assumed to independently reproduce in discrete and non-overlapping generation, $t = 0, 1, 2, \dots$. Thus, the number of offspring per allele (also an individual in this haploid population) at any generation is represented by a set of identically and individually distributed random variables. In particular, the numbers of offspring of M and W alleles (denoted as K_M and K_W respectively) can be represented by following distributions.

$$P(K_M = k) = i_k \text{ for } k = 0, 1, 2, \dots$$

$$P(K_W = k) = j_k \text{ for } k = 0, 1, 2, \dots$$

For a particular distribution, we can obtain the average number of offspring as follows.

$$E(K_M) = \sum_{k=0}^{\infty} k \times i_k$$

$$E(K_W) = \sum_{k=0}^{\infty} k \times j_k$$

With the selection coefficient of s (0 for neutral mutation, positive for advantageous mutation, and negative for deleterious mutation) of M allele, average offspring number of M allele and W allele will follow the relationship.

$$E(K_M) = E(K_W)(1 + s)$$

Now, the evolution of the number of M allele (denoted as M_t) and the number of W allele (denoted as W_t) as time process is a branching process.

To compare Haldane model with WF model (the offspring number follows Poisson distribution with variance and mean equal to 1), we will set $E(K_W) = 1$, $E(K_M) = 1 + s$. And then let the offspring number follows a more general and realistic distribution (including negative binomial distribution (negBin), beta-binomial distribution (betaBin)), which can let the ratio of variance to mean range from 1 to a large number. (For simplicity, we let $V(K_M)/E(K_M) = V(K_W)/E(K_W)$.) Based on the branching process, we obtained the fixation probability M alleles (see Supplementary Information for more details on the derivation).

$$P_f = \sum_{t=0}^{\infty} \left[1 - (u_M(t))^{M_0} \right] \left[(u_W(t))^{W_0} - (u_W(t-1))^{W_0} \right]$$

where

$$u_M(t) = P(M_t = 0 | M_0 = 1) = \sum_{k=0}^{\infty} P(M_t = 0 | M_{t-1} = k) = \sum_{k=0}^{\infty} i_k (u_M(t-1))^k$$

$$u_W(t) = P(W_t = 0 | W_0 = 1) = \sum_{k=0}^{\infty} P(W_t = 0 | W_{t-1} = k) = \sum_{k=0}^{\infty} j_k (u_W(t-1))^k$$

Note $u_M(t=1) = P(K_M = 0) = i_0$, $u_W(t=1) = P(K_W = 0) = j_0$. And both $\{u_M(t), t = 0, 1, 2, \dots\}$ and $\{u_W(t), t = 0, 1, 2, \dots\}$ are a bounded monotonic sequence. Thus, although we cannot obtain the direct analytical solution for P_f we could easily obtain their numerical solution by iteration to the case when both $u_M(t)$ and $u_W(t)$ converge.

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Competing interest statement

Authors declare that they have no competing interests.

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Editors

Reviewing Editor

Ziyue Gao

University of Pennsylvania, Philadelphia, United States of America

Senior Editor

George Perry

Pennsylvania State University, University Park, United States of America

Reviewer #1 (Public Review):

Summary:

The authors present a theoretical treatment of what they term the "Wright-Fisher-Haldane" model, a claimed modification of the standard model of genetic drift that accounts for variability in offspring number, and argue that it resolves a number of paradoxes in molecular evolution. Ultimately, I found this manuscript quite strange. The notion of effective population size as inversely related to the variance in offspring number is well known in the literature, and not exclusive to Haldane's branching process treatment. However, I found the authors' point about variance in offspring changing over the course of, e.g. exponential growth fairly interesting, and I'm not sure I'd seen that pointed out before. Nonetheless, I don't think the authors' modeling, simulations, or empirical data analysis are sufficient to justify their claims.

Weaknesses:

I have several outstanding issues. First of all, the authors really do not engage with the literature regarding different notions of an effective population. Most strikingly, the authors don't talk about Cannings models at all, which are a broad class of models with non-Poisson offspring distributions that nonetheless converge to the standard Wright-Fisher diffusion under many circumstances, and to "jumpy" diffusions/coalescents otherwise (see e.g. Mohle 1998, Sagitov (2003), Der et al (2011), etc.). Moreover, there is extensive literature on effective

population sizes in populations whose sizes vary with time, such as Sano et al (2004) and Sjodin et al (2005). Of course in many cases here the discussion is under neutrality, but it seems like the authors really need to engage with this literature more.

The most interesting part of the manuscript, I think, is the discussion of the Density Dependent Haldane model (DDH). However, I feel like I did not fully understand some of the derivation presented in this section, which might be my own fault. For instance, I can't tell if Equation 5 is a result or an assumption - when I attempted a naive derivation of Equation 5, I obtained $E(K_t) = 1 + r/c * (c-n) * dt$. It's unclear where the parameter z comes from, for example. Similarly, is equation 6 a derivation or an assumption? Finally, I'm not 100% sure how to interpret equation 7. Is that a variance effective size at time t ? Is it possible to obtain something like a coalescent N_e or an expected number of segregating sites or something from this?

Similarly, I don't understand their simulations. I expected that the authors would do individual-based simulations under a stochastic model of logistic growth, and show that you naturally get variance in offspring number that changes over time. But it seems that they simply used their equations 5 and 6 to fix those values. Moreover, I don't understand how they enforce population regulation in their simulations--is N_t random and determined by the (independent) draws from K_t for each individual? In that case, there's no "interaction" between individuals (except abstractly, since logistic growth arises from a model that assumes interactions between individuals). This seems problematic for their model, which is essentially motivated by the fact that early during logistic growth, there are basically no interactions, and later there are, which increases variance in reproduction. But their simulations assume no interactions throughout!

The authors also attempt to show that changing variance in reproductive success occurs naturally during exponential growth using a yeast experiment. However, the authors are not counting the offspring of individual yeast during growth (which I'm sure is quite hard). Instead, they use an equation that estimates the variance in offspring number based on the observed population size, as shown in the section "Estimation of $V(K)$ and $E(K)$ in yeast cells". This is fairly clever, however, I am not sure it is right, because the authors neglect covariance in offspring between individuals. My attempt at this derivation assumes that $I_t | I_{t-1} = \sum_{i=1}^{I_{t-1}} K_{i,t-1}$ where $K_{i,t-1}$ is the number of offspring of individual i at time $t-1$. Then, for example, $E(V(I_t | I_{t-1})) = E(V(\sum_{i=1}^{I_{t-1}} K_{i,t-1})) = E(I_{t-1})V(K_{t-1}) + E(I_{t-1}(I_{t-1}-1)) * Cov(K_{i,t-1}, K_{j,t-1})$. The authors have the first term, but not the second, and I'm not sure the second can be neglected (in fact, I believe it's the second term that's actually important, as early on during growth there is very little covariance because resources aren't constrained, but at carrying capacity, an individual having offspring means that another individual has to have fewer offspring - this is the whole notion of exchangeability, also neglected in this manuscript). As such, I don't believe that their analysis of the empirical data supports their claim.

Thus, while I think there are some interesting ideas in this manuscript, I believe it has some fundamental issues: first, it fails to engage thoroughly with the literature on a very important topic that has been studied extensively. Second, I do not believe their simulations are appropriate to show what they want to show. And finally, I don't think their empirical analysis shows what they want to show.

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Sjodin, P., et al. "On the meaning and existence of an effective population size." *Genetics* 169.2 (2005): 1061-1070

<https://doi.org/10.7554/eLife.99990.1.sa3>

Reviewer #2 (Public Review):

Summary:

This theoretical paper examines genetic drift in scenarios deviating from the standard Wright-Fisher model. The authors discuss Haldane's branching process model, highlighting that the variance in reproductive success equates to genetic drift. By integrating the Wright-Fisher model with the Haldane model, the authors derive theoretical results that resolve paradoxes related to effective population size.

Strengths:

The most significant and compelling result from this paper is perhaps that the probability of fixing a new beneficial mutation is $2s/V(K)$. This is an intriguing and potentially generalizable discovery that could be applied to many different study systems.

The authors also made a lot of effort to connect theory with various real-world examples, such as genetic diversity in sex chromosomes and reproductive variance across different species.

Weaknesses:

One way to define effective population size is by the inverse of the coalescent rate. This is where the geometric mean of N_e comes from. If N_e is defined this way, many of the paradoxes mentioned seem to resolve naturally. If we take this approach, one could easily show that a large N population can still have a low coalescent rate depending on the reproduction model. However, the authors did not discuss N_e in light of the coalescent theory. This is surprising given that Eldon and Wakeley's 2006 paper is cited in the introduction, and the multiple mergers coalescent was introduced to explain the discrepancy between census size and effective population size, superspreaders, and reproduction variance - that said, there is no explicit discussion or introduction of the multiple mergers coalescent.

The Wright-Fisher model is often treated as a special case of the Cannings 1974 model, which incorporates the variance in reproductive success. This model should be discussed. It is unclear to me whether the results here have to be explained by the newly introduced WFH model, or could have been explained by the existing Cannings model.

The abstract makes it difficult to discern the main focus of the paper. It spends most of the space introducing "paradoxes".

The standard Wright-Fisher model makes several assumptions, including hermaphroditism, non-overlapping generations, random mating, and no selection. It will be more helpful to clarify which assumptions are being violated in each tested scenario, as $V(K)$ is often not the only assumption being violated. For example, the logistic growth model assumes no cell death

at the exponential growth phase, so it also violates the assumption about non-overlapping generations.

The theory and data regarding sex chromosomes do not align. The fact that $\hat{\alpha}$ can be negative does not make sense. The authors claim that a negative $\hat{\alpha}$ is equivalent to infinity, but why is that? It is also unclear how θ is defined. It seems to me that one should take the first principle approach e.g., define θ as pairwise genetic diversity, and start with deriving the expected pair-wise coalescence time under the MMC model, rather than starting with assuming $\theta = 4N_e$. Overall, the theory in this section is not well supported by the data, and the explanation is insufficient.

<https://doi.org/10.7554/eLife.99990.1.sa2>

Reviewer #3 (Public Review):

Summary:

Ruan and colleagues consider a branching process model (in their terminology the "Haldane model") and the most basic Wright-Fisher model. They convincingly show that offspring distributions are usually non-Poissonian (as opposed to what's assumed in the Wright-Fisher model), and can depend on short-term ecological dynamics (e.g., variance in offspring number may be smaller during exponential growth). The authors discuss branching processes and the Wright-Fisher model in the context of 3 "paradoxes": (1) how N_e depends on N might depend on population dynamics; (2) how N_e is different on the X chromosome, the Y chromosome, and the autosomes, and these differences do match the expectations based on simple counts of the number of chromosomes in the populations; (3) how genetic drift interacts with selection. The authors provide some theoretical explanations for the role of variance in the offspring distribution in each of these three paradoxes. They also perform some experiments to directly measure the variance in offspring number, as well as perform some analyses of published data.

Strengths:

- (1) The theoretical results are well-described and easy to follow.
- (2) The analyses of different variances in offspring number (both experimentally and analyzing public data) are convincing that non-Poissonian offspring distributions are the norm.
- (3) The point that this variance can change as the population size (or population dynamics) change is also very interesting and important to keep in mind.
- (4) I enjoyed the Density-Dependent Haldane model. It was a nice example of the decoupling of census size and effective size.

Weaknesses:

- (1) I am not convinced that these types of effects cannot just be absorbed into some time-varying N_e and still be well-modeled by the Wright-Fisher process.
- (2) Along these lines, there is well-established literature showing that a broad class of processes (a large subset of Cannings' Exchangeable Models) converge to the Wright-Fisher diffusion, even those with non-Poissonian offspring distributions (e.g., Mohle and Sagitov 2001). E.g., equation (4) in Mohle and Sagitov 2001 shows that in such cases the "coalescent N_e " should be $(N-1) / \text{Var}(K)$, essentially matching equation (3) in the present paper.

(3) Beyond this, I would imagine that branching processes with heavy-tailed offspring distributions could result in deviations that are not well captured by the authors' WFH model. In this case, the processes are known to converge (backward-in-time) to Lambda or Xi coalescents (e.g., Eldon and Wakely 2006 or again in Mohle and Sagitov 2001 and subsequent papers), which have well-defined forward-in-time processes.

(4) These results that N_e in the Wright-Fisher process might not be related to N in any straightforward (or even one-to-one) way are well-known (e.g., Neher and Hallatschek 2012; Spence, Kamm, and Song 2016; Matuszewski, Hildebrandt, Achaz, and Jensen 2018; Rice, Novembre, and Desai 2018; the work of Lounès Chikhi on how N_e can be affected by population structure; etc...)

(5) I was also missing some discussion of the relationship between the branching process and the Wright-Fisher model (or more generally Cannings' Exchangeable Models) when conditioning on the total population size. In particular, if the offspring distribution is Poisson, then conditioned on the total population size, the branching process is identical to the Wright-Fisher model.

(6) In the discussion, it is claimed that the last glacial maximum could have caused the bottleneck observed in human populations currently residing outside of Africa. Compelling evidence has been amassed that this bottleneck is due to serial founder events associated with the out-of-Africa migration (see e.g., Henn, Cavalli-Sforza, and Feldman 2012 for an older review - subsequent work has only strengthened this view). For me, a more compelling example of changes in carrying capacity would be the advent of agriculture ~11kya and other more recent technological advances.

<https://doi.org/10.7554/eLife.99990.1.sa1>

Author response:

We thank you for the opportunity to provide a concise response. The criticisms are accurately summarized in the eLife assessment:

the study fails to engage prior literature that has extensively examined the impact of variance in offspring number, implying that some of the paradoxes presented might be resolved within existing frameworks.

The essence of our study is to propose the adoption of the Haldane model of genetic drift, based on the branching process, in lieu of the Wright-Fisher (WF) model, based on sampling, usually binomial. In addition to some extensions of the Haldane model, we present 4 paradoxes that cannot be resolved by the WF model. The reviews suggest that some of the paradoxes could be resolved by the WF model, if we engage prior literature sufficiently.

We certainly could not review all the literature on genetic drift as there must be thousands of them. Nevertheless, the literature we do not cover is based on the WF model, which has the general properties that all modifications of the WF model share. (We should note that all such modifications share the sampling aspect of the WF model. To model such sampling, N is imposed from outside of the model, rather than self-generating within the model. Most important, these modifications are mathematically valid but biologically untenable, as will be elaborated below. Thus, in concept, the WF and Haldane models are fundamentally different.)

In short, our proposal is general with the key point that the WF model cannot resolve these (and many other) paradoxes. The reviewers disagree (apparently only partially) and we shall

be specific in our response below.

We shall first present the 4th paradox, which is about multi-copy gene systems (such as rRNA genes and viruses, see the companion paper). Viruses evolve both within and between hosts. In both stages, there are severe bottlenecks. How does one address the genetic drift in viral evolution? How can we model the effective population sizes both within- and between- hosts? The inability of the WF model in dealing with such multi-copy gene systems may explain the difficulties in accounting for the SARS-CoV-2 evolution. Given the small number of virions transmitted between hosts, drift is strong which we have shown by using the Haldane model (Ruan, Luo, et al. 2021; Ruan, Wen, et al. 2021; Hou, et al. 2023).

As the reviewers suggest, it is possible to modify the WF model to account for some of these paradoxes. However, the modifications are often mathematically convenient but biologically dubious. Much of the debate is about the progeny number, K . (We shall use haploid model for this purpose but diploidy does not pose a problem as stated in the main text.) The modifications relax the constraint of $V(k) = E(k)$ inherent in the WF sampling. One would then ask how $V(k)$ can be different from $E(k)$ in the WF sampling even though it is mathematically feasible (but biologically dubious)? Kimura and Crow (1963) may be the first to offer a biological explanation. If one reads it carefully, Kimura's modification is to make the WF model like the Haldane model. Then, why don't we use the Haldane model in the first place by having two parameters, $E(k)$ and $V(k)$, instead of the one-parameter WF model?

The Haldane model is conceptually simpler. It allows the variation in population size, N , to be generated from within the model, rather than artificially imposed from outside of the model. This brings us to the first paradox, the density-dependent Haldane model. When N is increasing exponentially as in bacterial or yeast cultures, there is almost no drift when N is very low and drift becomes intense as N grows to near the carrying capacity. We do not see how the WF model can resolve this paradox, which can otherwise be resolved by the Haldane model.

The second and third paradoxes are about how much mathematical models of population genetic can be detached from biological mechanisms. The second paradox about sex chromosomes is rooted in the realization of $V(k) \neq E(k)$. Since $E(k)$ is the same between sexes but $V(k)$ is different, how does the WF sampling give rise to $V(k) \neq E(k)$? We are asking a biological question that troubled Kimura and Crow (1963) alluded above. The third paradox is acknowledged by two reviewers. Genetic drift manifested in the fixation probability of an advantageous mutation is $2s/V(k)$. It is thus strange that the fundamental parameter of drift in the WF model, N (or N_e), is missing. In the Haldane model, drift is determined by $V(k)$ with N being a scaling factor; hence $2s/V(k)$ makes perfect biological sense,

We now answer the obvious question: If the model is fundamentally about the Haldane model, why do we call it the WF-Haldane model? The reason is that most results obtained by the WF model are pretty good approximations and the branching process may not need to constantly re-derive the results. At least, one can use the WF results to see how well they fit into the Haldane model. In our earlier study (Chen, et al. (2017); Fig. 3), we show that the approximations can be very good in many (or most) settings.

We would like to use the modern analogy of gas-engine cars vs. electric-motor ones. The Haldane model and the WF model are as fundamentally different concepts as the driving mechanisms of gas-powered vs electric cars. The old model is now facing many problems and the fixes are often not possible. Some fixes are so complicated that one starts thinking about simpler solutions. The reservations are that we have invested so much in the old models which might be wasted by the switch. However, we are suggesting the integration of the WF and Haldane models. In this sense, the WF model has had many contributions which the new model gratefully inherits. This is true with the legacy of gas-engine cars inherited by EVs.

The editors also issue the instruction: while the modified model yields intriguing theoretical predictions, the simulations and empirical analyses are incomplete to support the authors' claims.

We are thankful to the editors and reviewers for the thoughtful comments and constructive criticisms. We also appreciate the publishing philosophy of eLife that allows exchanges, debates and improvements, which are the true spirits of science publishing.

References for the provisional author responses

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Review comments

eLife assessment

This study presents a useful modification of a standard model of genetic drift by incorporating variance in offspring numbers, claiming to address several paradoxes in molecular evolution.

It is unfortunate that the study fails to engage prior literature that has extensively examined the impact of variance in offspring number, implying that some of the paradoxes presented might be resolved within existing frameworks.

We do not believe that the paradoxes can be resolved.

In addition, while the modified model yields intriguing theoretical predictions, the simulations and empirical analyses are incomplete to support the authors' claims.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors present a theoretical treatment of what they term the "Wright-Fisher-Haldane" model, a claimed modification of the standard model of genetic drift that accounts for variability in offspring number, and argue that it resolves a number of paradoxes in molecular evolution. Ultimately, I found this manuscript quite strange.

The notion of effective population size as inversely related to the variance in offspring number is well known in the literature, and not exclusive to Haldane's branching process treatment. However, I found the authors' point about variance in offspring changing

over the course of, e.g. exponential growth fairly interesting, and I'm not sure I'd seen that pointed out before.

Nonetheless, I don't think the authors' modeling, simulations, or empirical data analysis are sufficient to justify their claims.

Weaknesses:

I have several outstanding issues. First of all, the authors really do not engage with the literature regarding different notions of an effective population. Most strikingly, the authors don't talk about Cannings models at all, which are a broad class of models with non-Poisson offspring distributions that nonetheless converge to the standard Wright-Fisher diffusion under many circumstances, and to "jumpy" diffusions/coalescents otherwise (see e.g. Mohle 1998, Sagitov (2003), Der et al (2011), etc.). Moreover, there is extensive literature on effective population sizes in populations whose sizes vary with time, such as Sano et al (2004) and Sjodin et al (2005).

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Similarly, I don't understand their simulations. I expected that the authors would do individual-based simulations under a stochastic model of logistic growth, and show that you naturally get variance in offspring number that changes over time. But it seems that they simply used their equations 5 and 6 to fix those values. Moreover, I don't understand how they enforce population regulation in their simulations---is N_t random and determined by the (independent) draws from K_t for each individual? In that case, there's no "interaction" between individuals (except abstractly, since logistic growth arises from a model that assumes interactions between individuals). This seems problematic for their model, which is essentially motivated by the fact that early during logistic growth, there are basically no interactions, and later there are, which increases variance in reproduction. But their simulations assume no interactions throughout!

The authors also attempt to show that changing variance in reproductive success occurs naturally during exponential growth using a yeast experiment. However, the authors are not counting the offspring of individual yeast during growth (which I'm sure is quite hard). Instead, they use an equation that estimates the variance in offspring number based on the observed population size, as shown in the section "Estimation of $V(K)$ and $E(K)$ in yeast cells". This is fairly clever, however, I am not sure it is right, because the authors neglect covariance in offspring between individuals. My attempt at this derivation assumes that $I_t \mid I_{t-1} = \sum_{i=1}^{I_{t-1}} K_{i,t-1}$ where $K_{i,t-1}$ is the number of offspring of individual i at time $t-1$. Then, for example, $E(V(I_t \mid I_{t-1})) = E(V(\sum_{i=1}^{I_{t-1}} K_{i,t-1})) = E(I_{t-1})V(K_{t-1}) + E(I_{t-1}(I_{t-1}-1))*Cov(K_{i,t-1}, K_{j,t-1})$. The authors have the first term, but not the second, and I'm not sure the second can be neglected (in fact, I believe it's the second term that's actually important, as early on during growth there is very little covariance because resources aren't constrained, but at carrying capacity, an individual having offspring means that another

individuals has to have fewer offspring - this is the whole notion of exchangeability, also neglected in this manuscript). As such, I don't believe that their analysis of the empirical data supports their claim.

Thus, while I think there are some interesting ideas in this manuscript, I believe it has some fundamental issues:

first, it fails to engage thoroughly with the literature on a very important topic that has been studied extensively. Second, I do not believe their simulations are appropriate to show what they want to show. And finally, I don't think their empirical analysis shows what they want to show.

References:

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Sjodin, P., et al. "On the meaning and existence of an effective population size." *Genetics* 169.2 (2005): 1061-1070

Reviewer #2 (Public Review):

Summary:

This theoretical paper examines genetic drift in scenarios deviating from the standard Wright-Fisher model. The authors discuss Haldane's branching process model, highlighting that the variance in reproductive success equates to genetic drift. By integrating the Wright-Fisher model with the Haldane model, the authors derive theoretical results that resolve paradoxes related to effective population size.

Strengths:

The most significant and compelling result from this paper is perhaps that the probability of fixing a new beneficial mutation is $2s/V(K)$. This is an intriguing and potentially generalizable discovery that could be applied to many different study systems.

The authors also made a lot of effort to connect theory with various real-world examples, such as genetic diversity in sex chromosomes and reproductive variance across different species.

Weaknesses:

One way to define effective population size is by the inverse of the coalescent rate. This is where the geometric mean of N_e comes from. If N_e is defined this way, many of the paradoxes mentioned seem to resolve naturally. If we take this approach, one could easily show that a large N population can still have a low coalescent rate depending on the reproduction model. However, the authors did not discuss N_e in light of the coalescent theory. This is surprising given that Eldon and Wakeley's 2006 paper is cited in the introduction, and the multiple mergers coalescent was introduced to explain the discrepancy between census size and effective population size, superspreaders, and

reproduction variance - that said, there is no explicit discussion or introduction of the multiple mergers coalescent.

The Wright-Fisher model is often treated as a special case of the Cannings 1974 model, which incorporates the variance in reproductive success. This model should be discussed. It is unclear to me whether the results here have to be explained by the newly introduced WFH model, or could have been explained by the existing Cannings model.

The abstract makes it difficult to discern the main focus of the paper. It spends most of the space introducing "paradoxes".

The standard Wright-Fisher model makes several assumptions, including hermaphroditism, non-overlapping generations, random mating, and no selection. It will be more helpful to clarify which assumptions are being violated in each tested scenario, as $V(K)$ is often not the only assumption being violated. For example, the logistic growth model assumes no cell death at the exponential growth phase, so it also violates the assumption about non-overlapping generations.

The theory and data regarding sex chromosomes do not align. The fact that $\hat{\alpha}$ can be negative does not make sense. The authors claim that a negative $\hat{\alpha}$ is equivalent to infinity, but why is that? It is also unclear how θ is defined. It seems to me that one should take the first principle approach e.g., define θ as pairwise genetic diversity, and start with deriving the expected pair-wise coalescence time under the MMC model, rather than starting with assuming $\theta = 4N_e$. Overall, the theory in this section is not well supported by the data, and the explanation is insufficient.

$\{\alpha$ and α' can both be negative. $X^2 = 0.47$ would yield $x = -0.7\}$

Reviewer #3 (Public Review):

Summary:

Ruan and colleagues consider a branching process model (in their terminology the "Haldane model") and the most basic Wright-Fisher model. They convincingly show that offspring distributions are usually non-Poissonian (as opposed to what's assumed in the Wright-Fisher model), and can depend on short-term ecological dynamics (e.g., variance in offspring number may be smaller during exponential growth). The authors discuss branching processes and the Wright-Fisher model in the context of 3 "paradoxes": (1) how N_e depends on N might depend on population dynamics; (2) how N_e is different on the X chromosome, the Y chromosome, and the autosomes, and these differences do match the expectations based on simple counts of the number of chromosomes in the populations; (3) how genetic drift interacts with selection. The authors provide some theoretical explanations for the role of variance in the offspring distribution in each of these three paradoxes. They also perform some experiments to directly measure the variance in offspring number, as well as perform some analyses of published data.

Strengths:

- (1) The theoretical results are well-described and easy to follow.
- (2) The analyses of different variances in offspring number (both experimentally and analyzing public data) are convincing that non-Poissonian offspring distributions are the norm.
- (3) The point that this variance can change as the population size (or population dynamics) change is also very interesting and important to keep in mind.

(4) I enjoyed the Density-Dependent Haldane model. It was a nice example of the decoupling of census size and effective size.

Weaknesses:

(1) I am not convinced that these types of effects cannot just be absorbed into some time-varying N_e and still be well-modeled by the Wright-Fisher process.

(2) Along these lines, there is well-established literature showing that a broad class of processes (a large subset of Cannings' Exchangeable Models) converge to the Wright-Fisher diffusion, even those with non-Poissonian offspring distributions (e.g., Mohle and Sagitov 2001). E.g., equation (4) in Mohle and Sagitov 2001 shows that in such cases the "coalescent N_e " should be $(N-1) / \text{Var}(K)$, essentially matching equation (3) in the present paper.

(3) Beyond this, I would imagine that branching processes with heavy-tailed offspring distributions could result in deviations that are not well captured by the authors' WFH model. In this case, the processes are known to converge (backward-in-time) to Lambda or Xi coalescents (e.g., Eldon and Wakely 2006 or again in Mohle and Sagitov 2001 and subsequent papers), which have well-defined forward-in-time processes.

(4) These results that N_e in the Wright-Fisher process might not be related to N in any straightforward (or even one-to-one) way are well-known (e.g., Neher and Hallatschek 2012; Spence, Kamm, and Song 2016; Matuszewski, Hildebrandt, Achaz, and Jensen 2018; Rice, Novembre, and Desai 2018; the work of Lounès Chikhi on how N_e can be affected by population structure; etc...)

(5) I was also missing some discussion of the relationship between the branching process and the Wright-Fisher model (or more generally Cannings' Exchangeable Models) when conditioning on the total population size. In particular, if the offspring distribution is Poisson, then conditioned on the total population size, the branching process is identical to the Wright-Fisher model.

(6) In the discussion, it is claimed that the last glacial maximum could have caused the bottleneck observed in human populations currently residing outside of Africa. Compelling evidence has been amassed that this bottleneck is due to serial founder events associated with the out-of-Africa migration (see e.g., Henn, Cavalli-Sforza, and Feldman 2012 for an older review - subsequent work has only strengthened this view). For me, a more compelling example of changes in carrying capacity would be the advent of agriculture ~11kya and other more recent technological advances.

Recommendations for the authors:

Reviewing Editor Comments:

The reviewers recognize the value of this model and some of the findings, particularly results from the density-dependent Haldane model. However, they expressed considerable concerns with the model and overall framing of this manuscript.

First, all reviewers pointed out that the manuscript does not sufficiently engage with the extensive literature on various models of effective population size and genetic drift, notably lacking discussion on Cannings models and related works.

Second, there is a disproportionate discussion on the paradoxes, yet some of the paradoxes might already be resolved within current theoretical frameworks. All three reviewers found the modeling and simulation of the yeast growth experiment hard to

follow or lacking justification for certain choices. The analysis approach of sex chromosomes is also questioned.

The reviewers recommend a more thorough review of relevant prior literature to better contextualize their findings. The authors need to clarify and/or modify their derivations and simulations of the yeast growth experiment to address the identified caveats and ensure robustness. Additionally, the empirical analysis of the sex chromosome should be revisited, considering alternative scenarios rather than relying solely on the MSE, which only provides a superficial solution. Furthermore, the manuscript's overall framing should be adjusted to emphasize the conclusions drawn from the WFH model, rather than focusing on the "unresolved paradoxes", as some of these may be more readily explained by existing frameworks. Please see the reviewers' overall assessment and specific comments.

Reviewer #2 (Recommendations For The Authors):

In the introduction -- "Genetic drift is simply $V(K)$ " -- this is a very strong statement. You can say it is inversely proportional to $V(K)$, but drift is often defined based on changes in allele frequency.

Page 3 line 86. "sexes is a sufficient explanation."--> "sex could be a sufficient explanation"

The strongest line of new results is about $2s/V(K)$. Perhaps, the paper could put more emphasis on this part and demonstrate the generality of this result with a different example.

The math notations in the supplement are not intuitive. e.g., using i_k and j_k as probabilities. I also recommend using $E[X]$ and $V[X]$ for expectation and variance rather than $\text{italic}\{E(X)\}$ to improve the readability of many equations.

Eq A6, A7, While I manage to follow, $P_{10}(t)$ and P_{10} are not defined anywhere in the text.

Supplement page 7, the term "probability of fixation" is confusing in a branching model.

E.q. A 28. It is unclear eq. A.1 could be used here directly. Some justification would be nice.

Supplement page 17. "the biological meaning of negative..". There is no clear justification for this claim. As a reader, I don't have any intuition as to why that is the case.

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